

Time: 2hrs	DIVISION OF MECHANICAL ENGINEERING, SoE, CUSAT ME19-200-0203 ENGINEERING MECHANICS (ME-A) Internal Test-2 (July 2024)	Marks:50
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Course Outcomes (CO):

On completion of the course, the student will be able to

CO1: Understand the principles of mechanics (statics and dynamics), the concept of free body diagrams, and resolution of forces.

CO2: Apply the principles of mechanics, concept of free body diagrams, resolution of forces, and equations of equilibrium or motion to a given engineering or physical application.

CO3: Analyse a given engineering problem or physical applications and **calculate** the required parameters like forces, moments, displacement, velocity, acceleration, etc.

CO4: Appreciate the physical and mathematical meaning of quantities like centroid and moment of inertia and their applications in engineering and **locate** the centroid and **calculate** the moment of inertia of different sections used in different engineering applications.

Bloom's Taxonomy Levels (BL)

Remember	Understand	Apply	Analyse	Design	Create
L1	L2	L3	L4	L5	L6

Programme Outcomes (PO)

PO1-Engineering Knowledge	PO2-Problem Analysis	PO3-Development of solutions	PO4- Conduct investigation of complex problems
PO5- Modern tool usage	PO6- Engineer& Society	PO7-Environment & Sustainability	PO8-Ethics
PO9-Individual& team work	PO10-Communication	PO11-Project management &finance	PO12-Lifelong learning

Programme Educational Objectives (PEO)

PEO1	Preparation: To prepare the graduates or to succeed in industry/technical profession through global rigorous education.
PEO2	Core Competence: To provide the graduates with a solid foundation in mathematical, scientific, and engineering fundamentals required to solve engineering problems and also to pursue higher studies
PEO3	Breadth: To train the graduates with good scientific and engineering breadth so as to comprehend, analyse, design, and create novel solutions and products for the real-life problems
PEO4	Professionalism: To inculcate in the graduates, a professional and ethical attitude, effective communication and team work skills, a multi-disciplinary approach, and an ability to relate engineering issues to broader social context
PEO5	Learning environment: To provide the graduates with an academic environment making them aware of excellence, leadership and teamwork skills, ethical codes and guidelines, and life-long learning needed for a successful professional career

Programme Specific Outcomes (PSO)

PSO1	The graduate should excel in product design, thermal and fluid engineering, and manufacturing systems with the acquired knowledge in mathematics, science, and engineering disciplines.
PSO2	The graduate should be competent to analyze, interpret, and solve real life mechanical engineering problems
PSO3	The graduate should have the managerial skills to work effectively in a team and in a society by following ethical and environmental practices

Mapping of Course Outcomes with Programme Outcomes and Programme Specific Outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	1	1				1				1	3	3	
CO2	3	3	1	3				1				1	3	3	
CO3	3	3	1	3				1				1	3	3	
CO4	3	3	1	1				1				1	3	3	

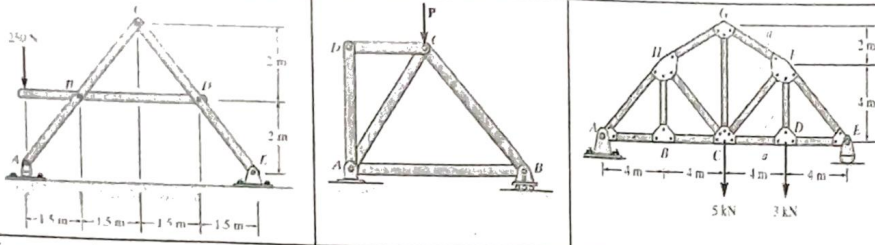
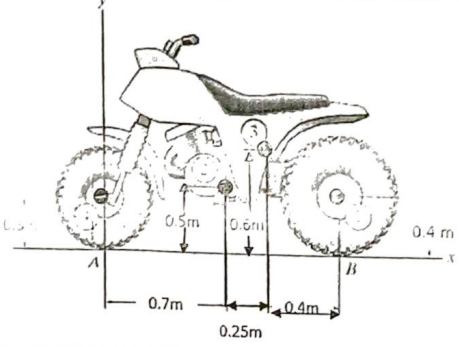
Mapping of Questions with Course Outcomes

	CO1	CO2	CO3	CO4	
Q1	2		3	5	
Q2	2	3	3		
Q3	2	3	3		
Q4				8	
Q5		4	4		
Q6		4	4		
Total	6	14	17	13	
%	12%	28%	34%	26%	

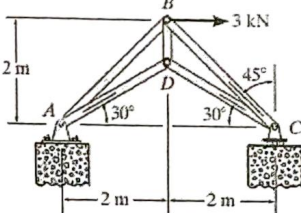
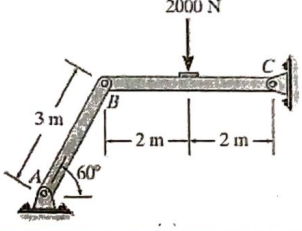
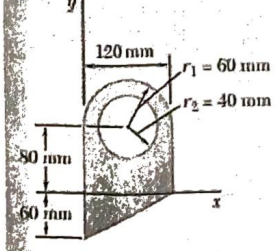
Mapping of Questions with Blooms Taxonomy levels

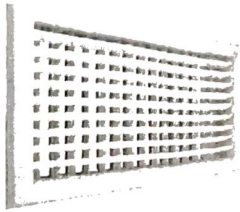
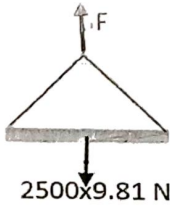
	L1	L2	L3	L4	L5	L6	
	2	3	5				
				8			
				8			
			8				
				8			
				8			
	2	3	13	32			
	4%	6%	26%	64%			

PART A

No	Question	CO	BL	Marks
a	State and explain D'Alemberts' Principle in rectilinear translation	CO1	L1	2
b	Identify trusses and frames from the figures given below and justify your answer.			
		CO3	L2	3
c	Locate the centre of gravity of the two-wheeler shown. The weights of each component are as given below and their respective centre of gravity is shown in the figure. 1. Rear wheel: 80 N 2. Engine assembly: 378 N 3. Frame: 530 N 4. Front wheel: 36 N			
		CO4	L3	5

PART B

No	Question	CO	BL	Marks
2	Determine the forces in the bars of the truss shown			
		CO1		2
		CO2	L4	3
		CO3		3
3	Determine the forces experienced by the pins at A, B, and C of the frame shown.			
		CO1		2
		CO2	L4	3
		CO3		3
4	Locate the centroid of the shaded area with respect to the co-ordinate axes			
		CO4	L3	8

5.	A louver vent is a kind of physical installation that helps vent air through a fixture that incorporates slanted pieces for natural control in heating and cooling systems. Experimental data indicate that in a region downstream of a given louvered supply vent, the velocity of the emitted air is defined by $v = \frac{0.18v_0}{x}$, where v and x are expressed in m/s and m respectively, and v_0 is the initial discharge velocity of air. For $v_0 = 3.6 \text{ m/s}$, determine i) Acceleration of the air at $x=2\text{m}$. The time required for the air to flow from $x=1$ to $x=3 \text{ m}$ 	CO2	L4	4
		CO3		4
6.	For a short time, the crane shown in figure lifts the 2500 kg beam with a force of $F = (28 + 3s^2) \times 10^3 \text{ N}$. Determine the speed of the beam when it has risen $s = 3\text{m}$. How much time does it take to attain this height starting from rest?  	CO2	L4	4
		CO3		4